Landau Singularities: Scope and Applications to QCD

Ajay S. Sakthivasan

[1]HISKP & BCTP, University of Bonn

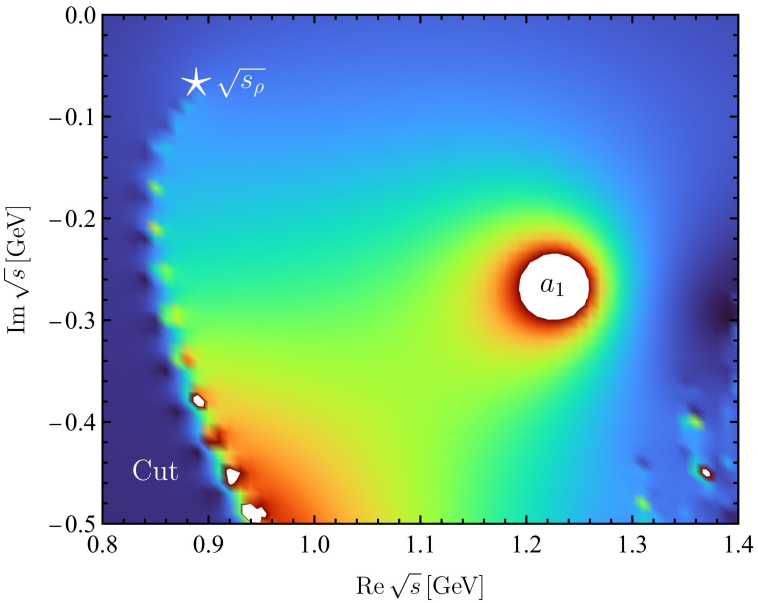
July 3, 2025

Presentation Outline

1. Introduction
2. Analytic Aspects
3. Infinite Volume Unitarity Formalism
4. Conclusions

Introduction

- Hadronic resonances correspond to poles of the S-matrix.

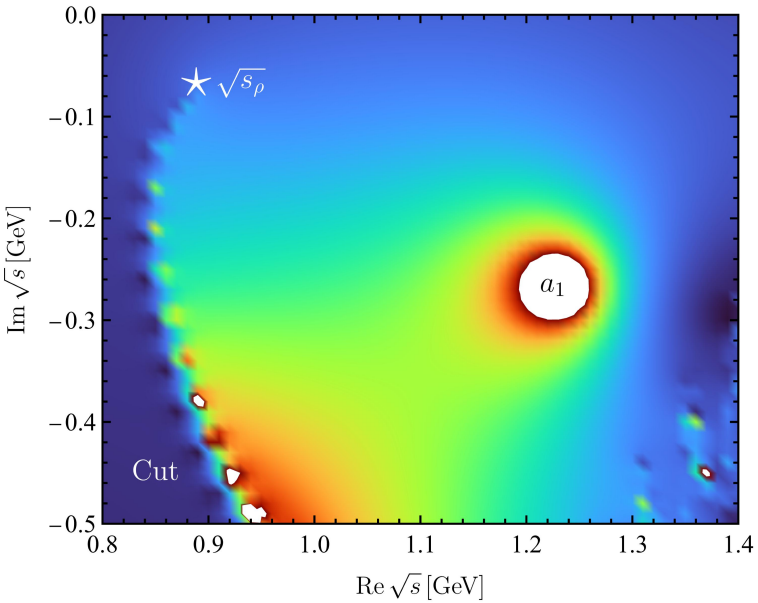


Pole position of the $a_1(1260)$.[1]

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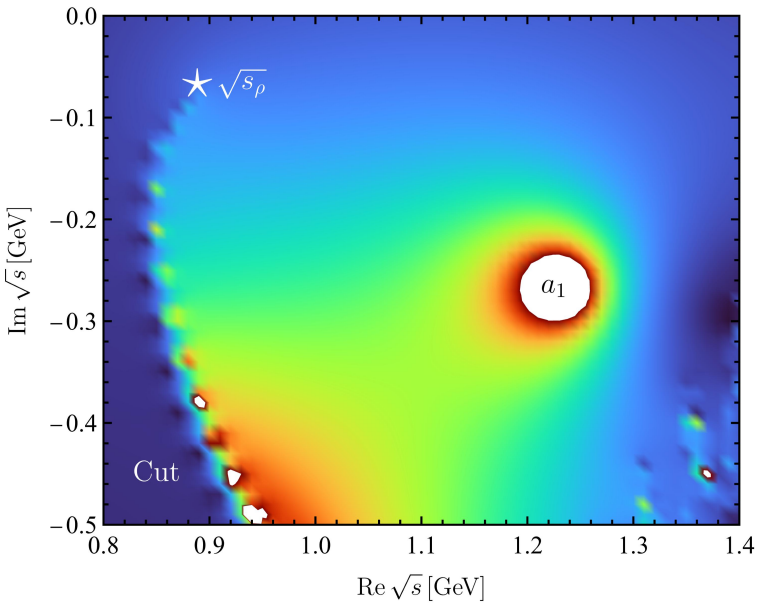


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- Experimentally — peaks in invariant mass distributions.
- All observed peaks correspond to genuine resonances? No — dynamically generated resonances, kinematic singularities, etc.



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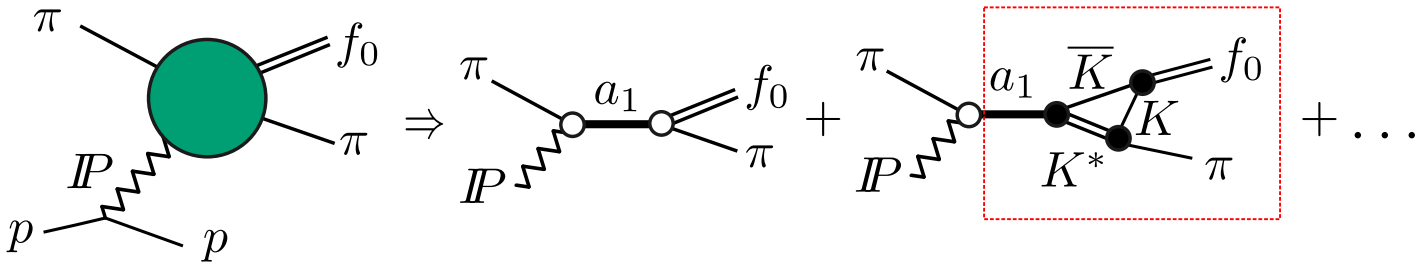
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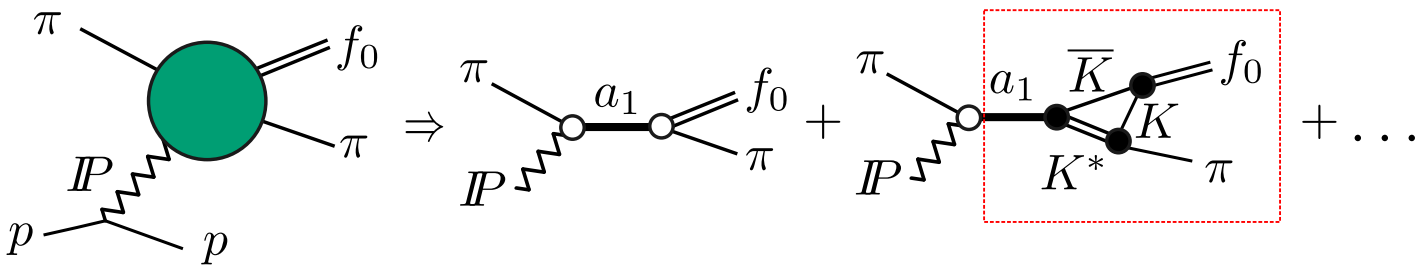
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- Effects of Final State Interactions on Landau singularities (AS, Mai, Rusetsky & Döring, JHEP10(2024)).

Effects of final state interactions on Landau singularities

A.S. Sakthivasan [a], M. Mai [a,b,c], A. Rusetsky [a,d] and M. Döring [c,e]

[a] *Helmholtz-Institut für Strahlen- und Kernphysik (Theorie) and Bethe Center for Theoretical Physics, Universität Bonn, 53115 Bonn, Germany*

[b] *Albert Einstein Center for Fundamental Physics, Institute for Theoretical Physics, University of Bern, Sidlerstrasse 5, 3012 Bern, Switzerland*

[c] *Institute for Nuclear Studies and Department of Physics, The George Washington University, Washington, DC 20052, U.S.A.*

[d] *Tbilisi State University, 0186 Tbilisi, Georgia*

[e] *Theory Center, Thomas Jefferson National Accelerator Facility, Newport News, VA 23606, U.S.A.*

E-mail: sakthivasan@hiskp.uni-bonn.de, mai@hiskp.uni-bonn.de, rusetsky@hiskp.uni-bonn.de, doring@gwu.edu

Abstract: In certain kinematic and particle mass configurations, triangle singularities may lead to line-shapes which mimic the effects of resonances. This well-known effect is scrutinized here in the presence of final-state rescattering. The goal is achieved first by utilizing general arguments provided by Landau equations, and second by applying a modern scattering formalism with explicit two- and three-body unitarity.

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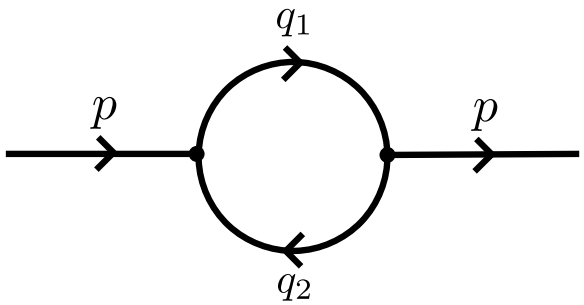
- For some real parameters α_i, the Landau equations read,

$$\alpha_i (q_i^2 - m_i^2) = 0 \implies \alpha_i = 0 \text{ or } q_i^2 = m_i^2,$$
$$\sum_{i \in \text{loop}} \alpha_i \; q_i^\mu(k_j) = 0.$$

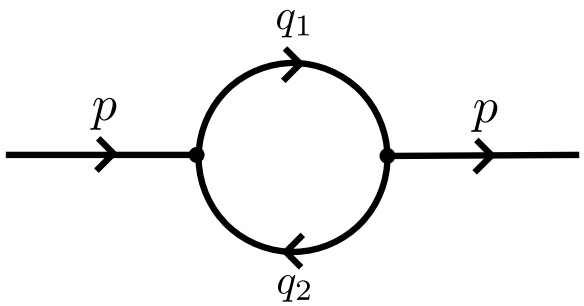
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- $\alpha_i = 0$ case is called the subleading singularity — equivalent to contracting the propagator.

- As an example, the bubble graph:



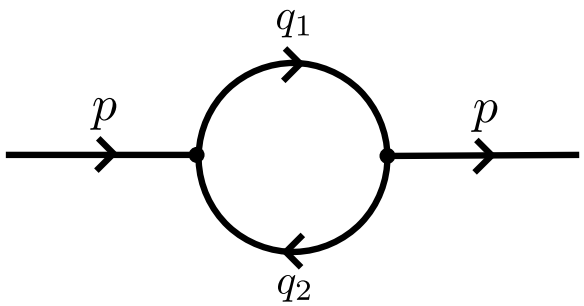
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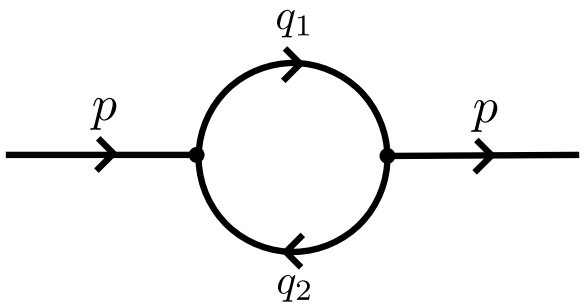


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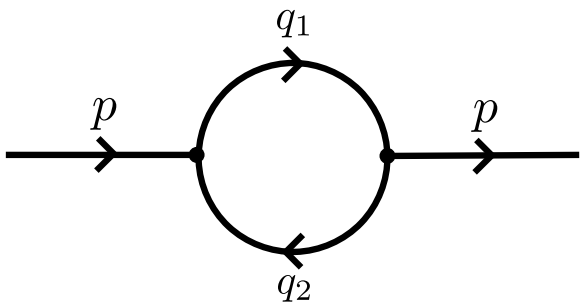


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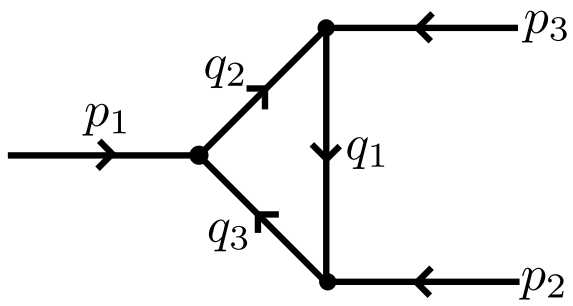


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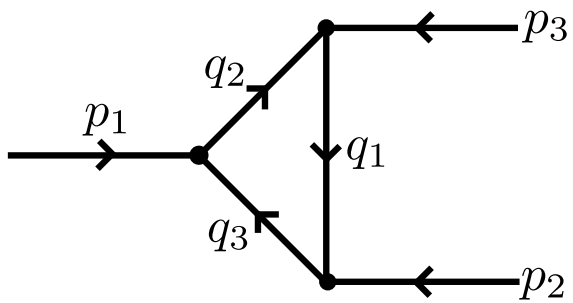
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- It is also possible to determine the nature of the singularity — in this case, it is a square root branch point.

- The triangle graph and the box graph are of relevance to us.

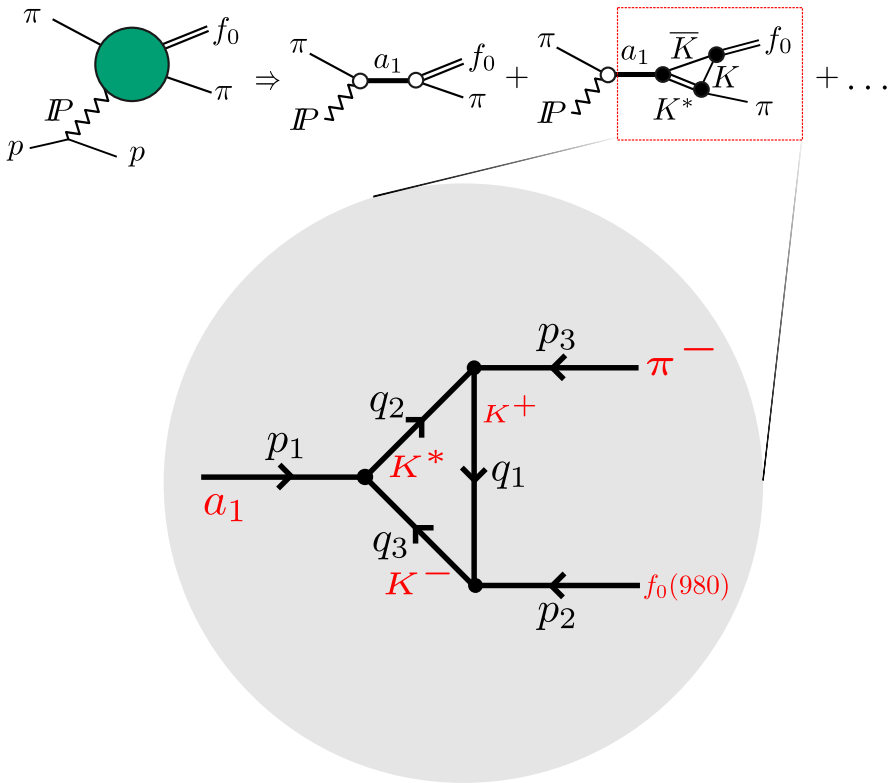


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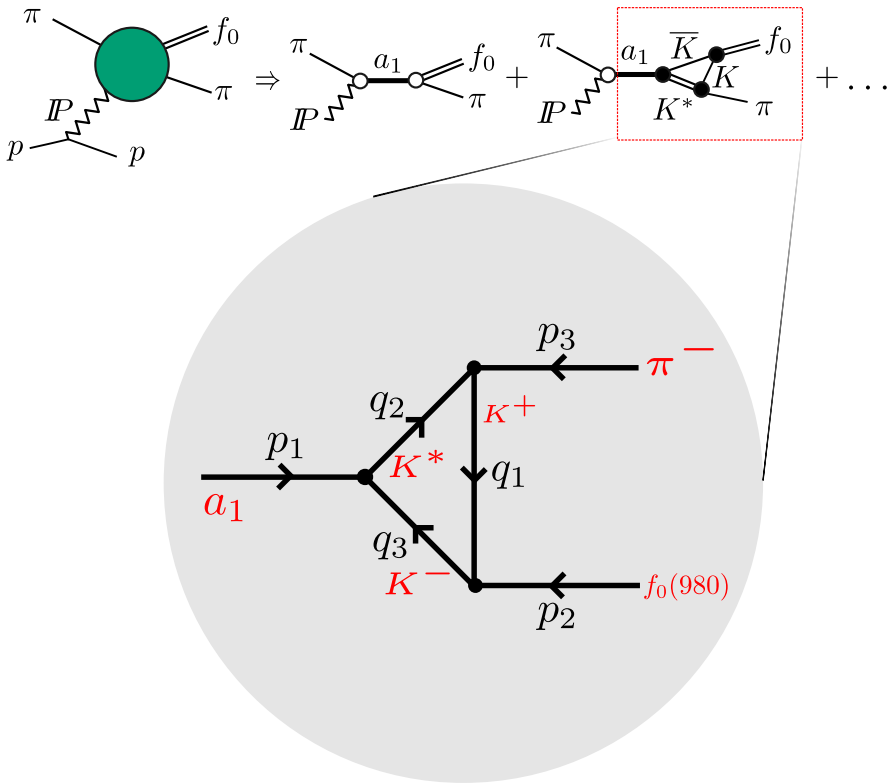
- These are more interesting, as they have more degrees of freedom; And also more singularities.

- The leading Landau singularity associated with the graph:



- $K*(892)$ is a vector resonance, $f_0(980)$ is a scalar resonance.

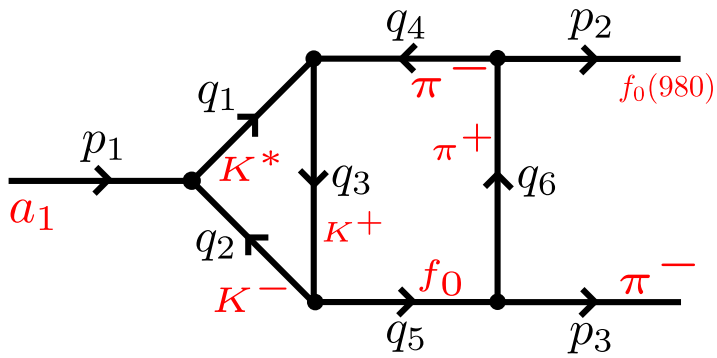
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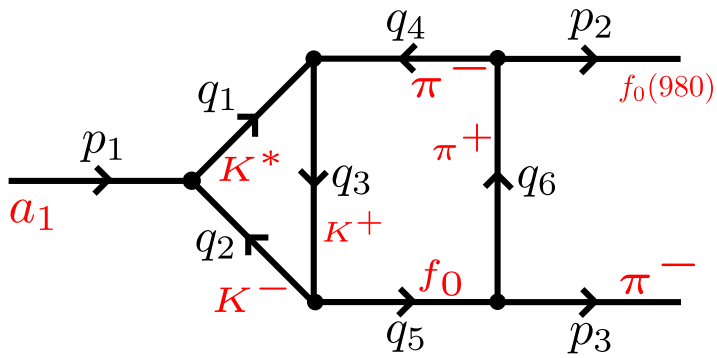
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- Landau singularity for $p_3 = m_\pi$: $p_2/\mathrm{GeV} \equiv \sqrt{\sigma}/\mathrm{GeV} \in [0.986, 1.024]$ and $p_1/\mathrm{GeV} \equiv \sqrt{s}/\mathrm{GeV} \in [1.385, 1.436]$.

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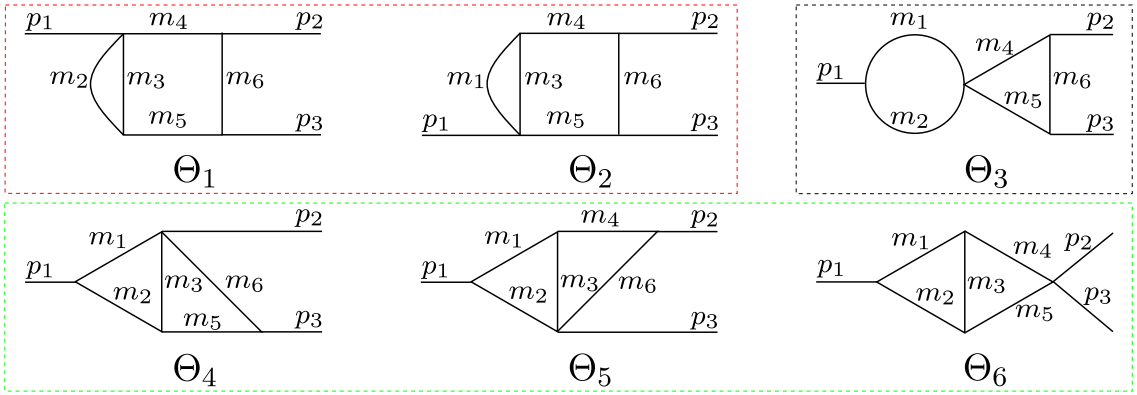


- Check for leading and subleading singularities. . .

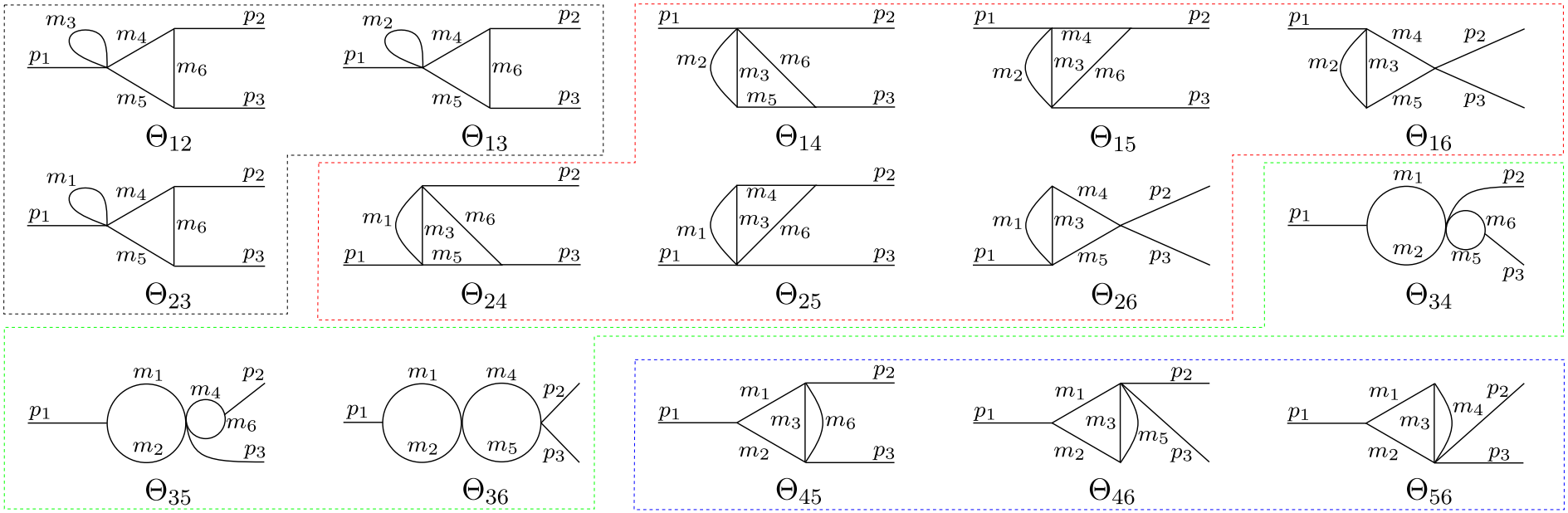
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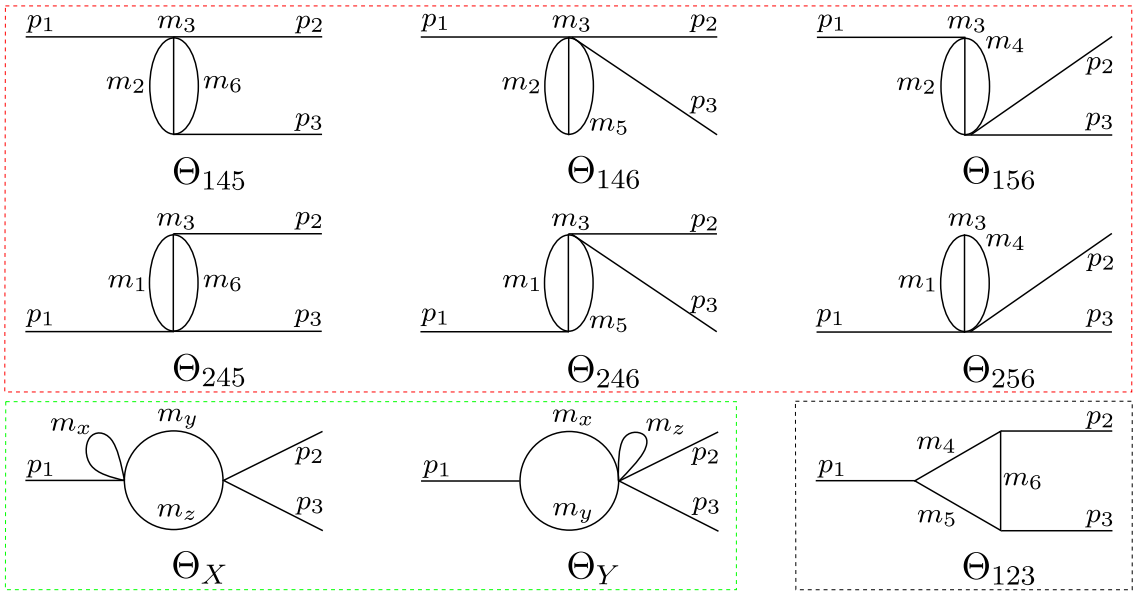
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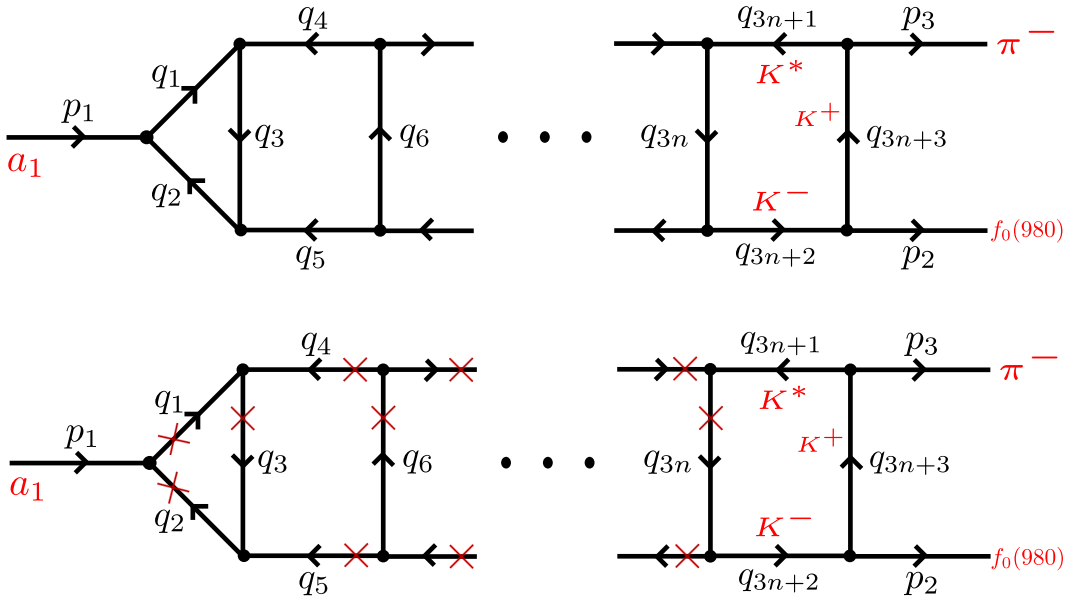


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- The triangle singularity is present as the subleading singularity; And no other singularities.



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- One still needs to evaulate the corresponding Feynman integrals.

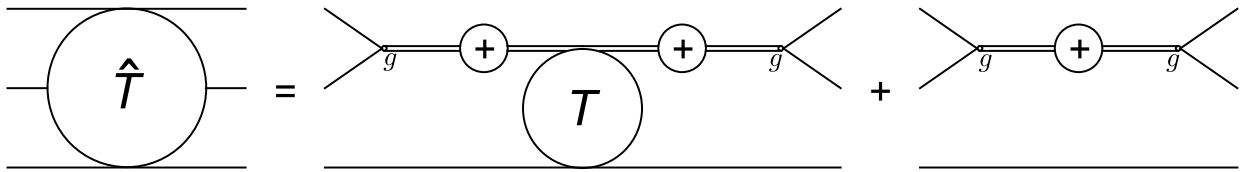
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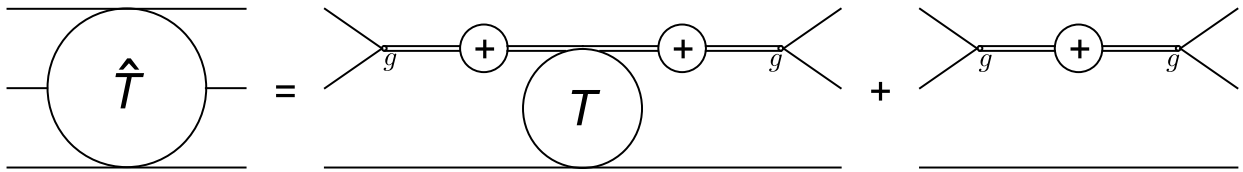
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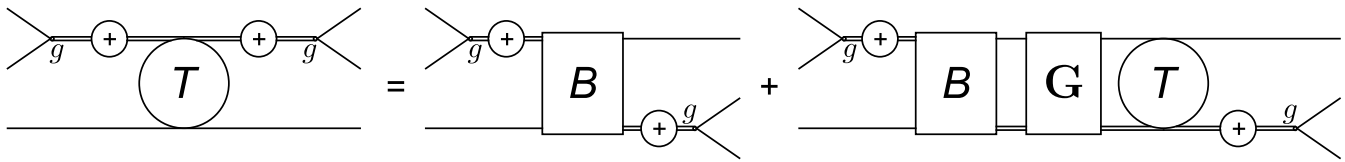
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- One considers the Bethe-Salpeter ansatz, $T = B + BGT$, to determine T.[4]



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- The equation to be solved,

$$T(p,q) = B(p,q) + \int \frac{d^3\mathbf{l}}{(2\pi)^3} \frac{1}{2E_l} B(p,l)\tau(\sigma(l))T(l,q).$$

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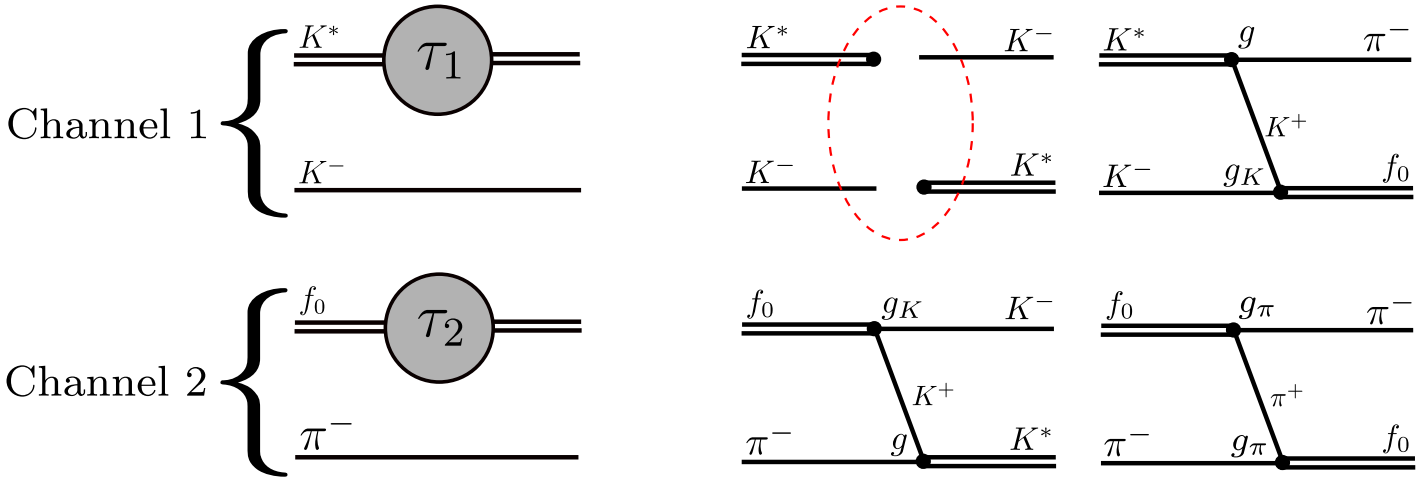
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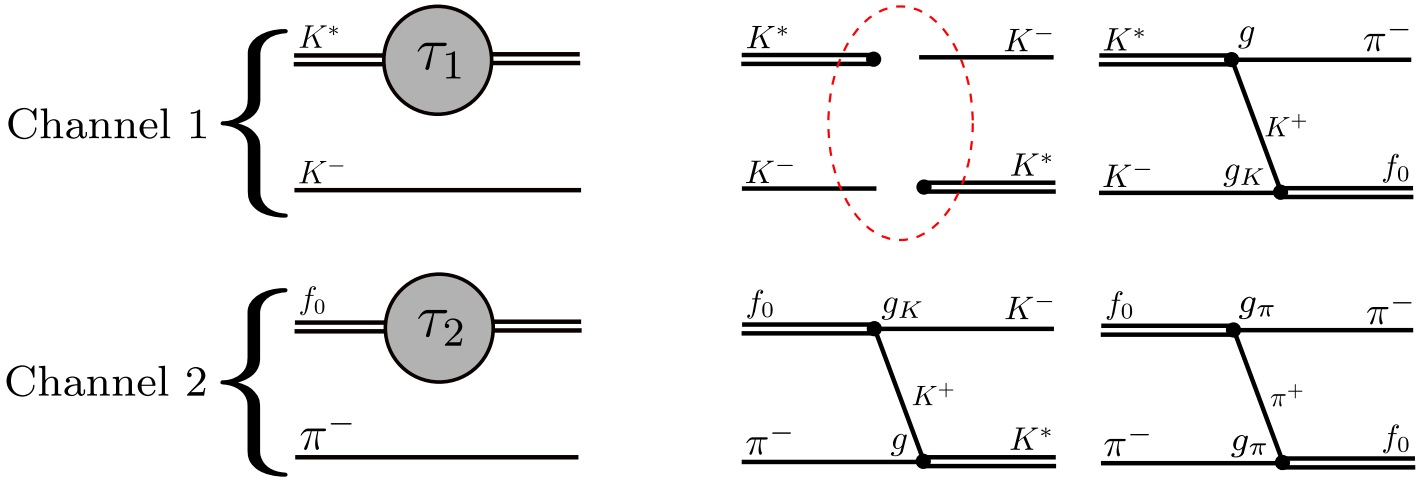
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- In this work, we consider scalar propagators, in relative s-wave.

- Need to carry out a coupled channel analysis.

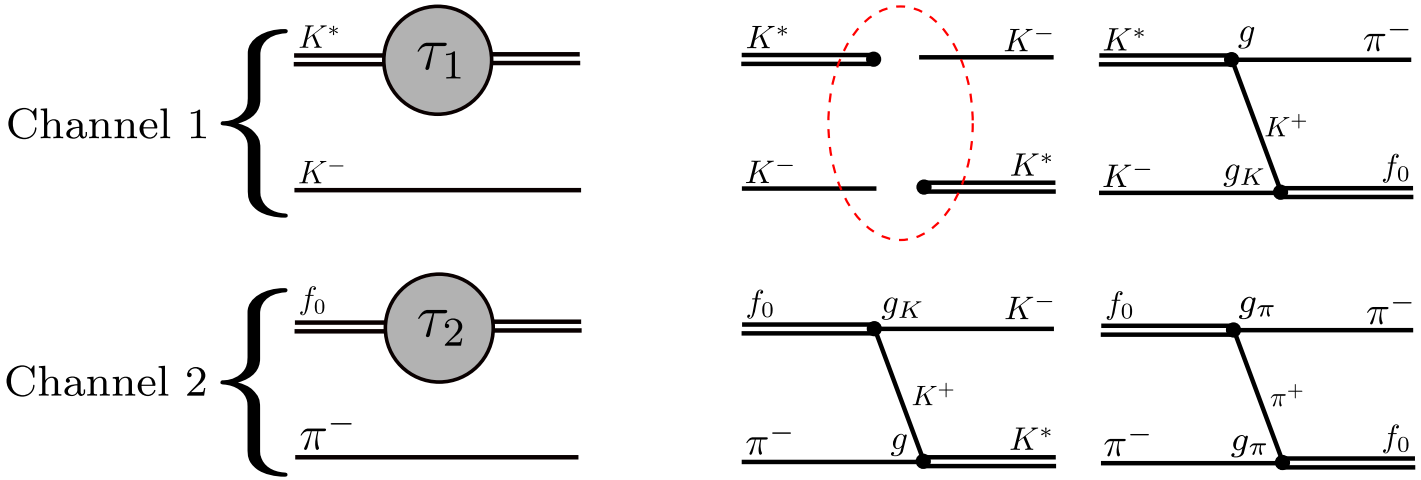


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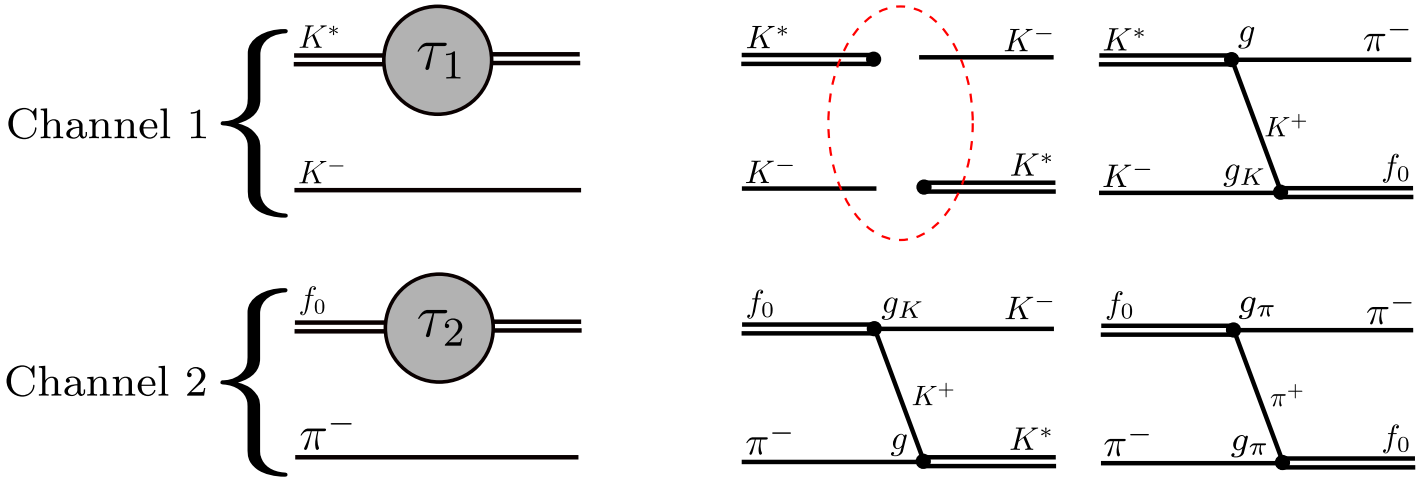
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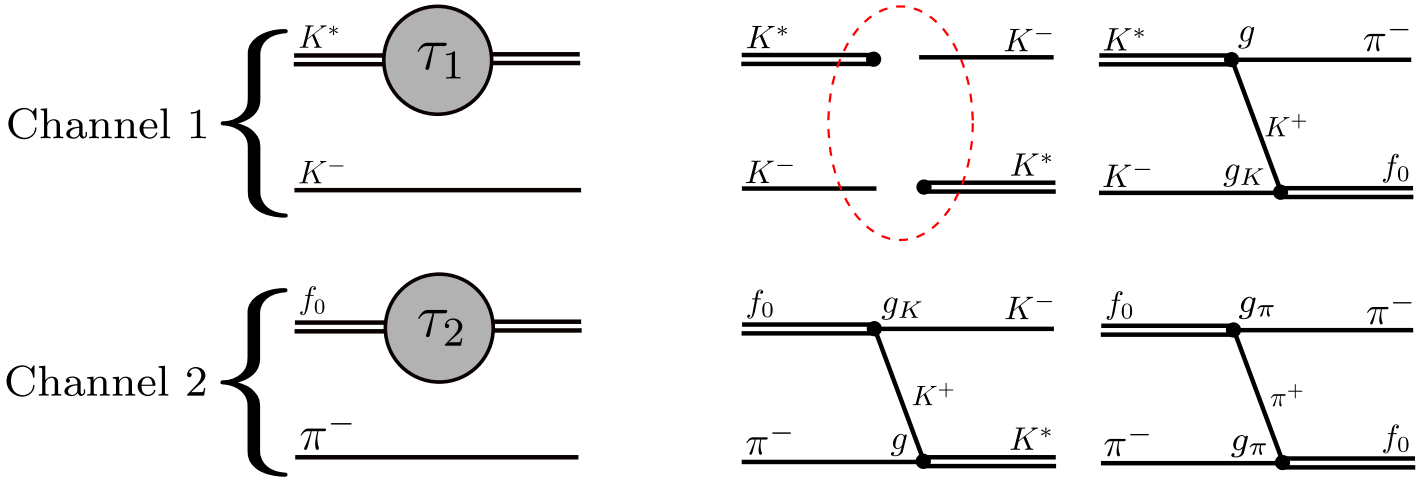
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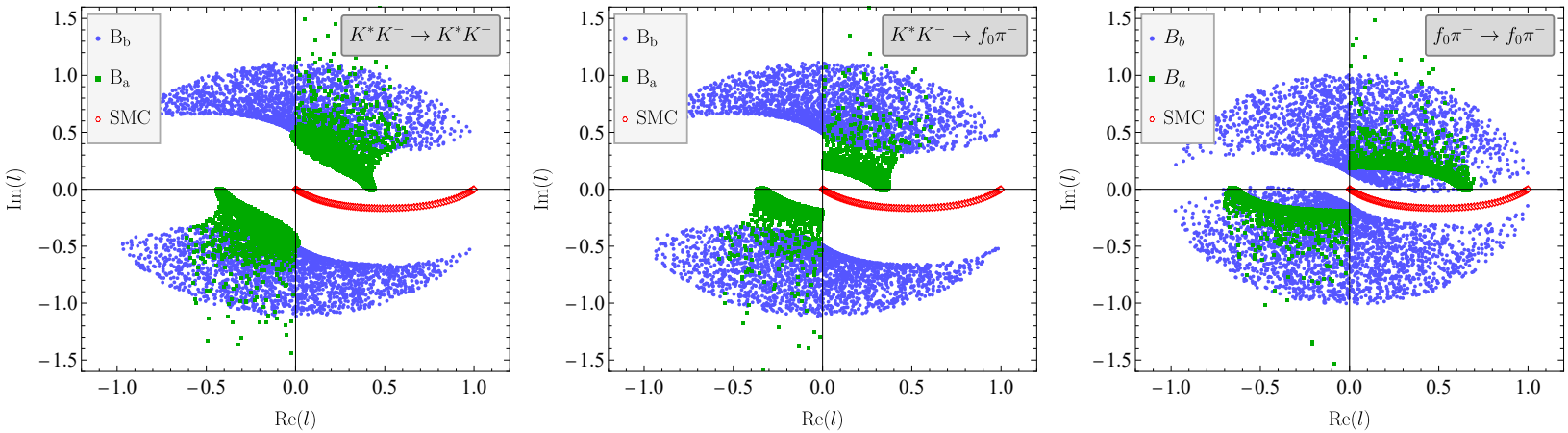


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- For f_0 we also make use of g_K/g_p ratio from BaBaR collaboration.

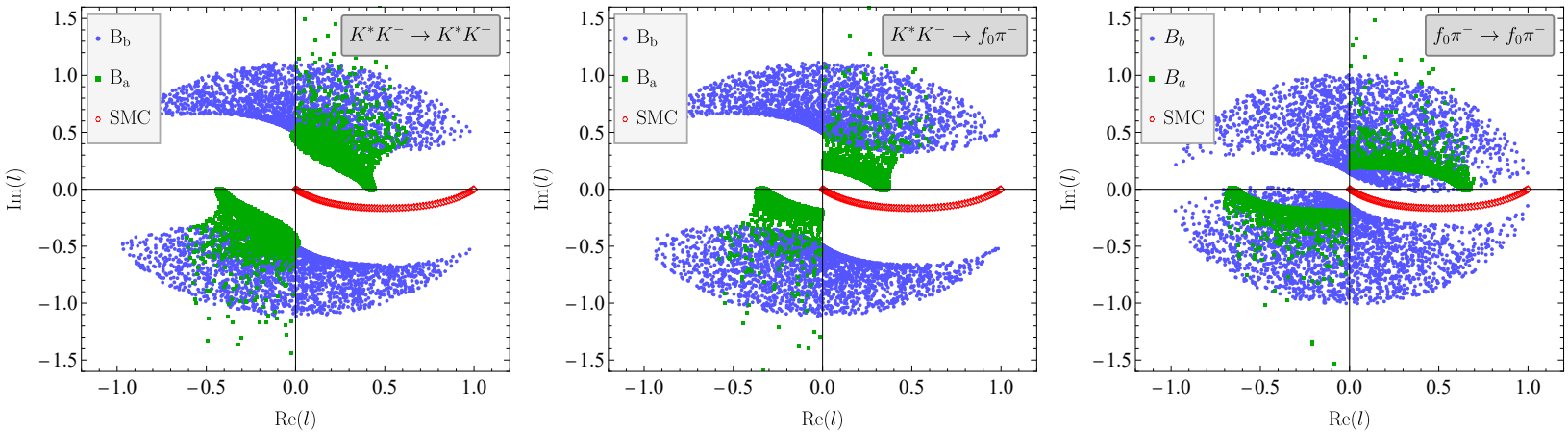
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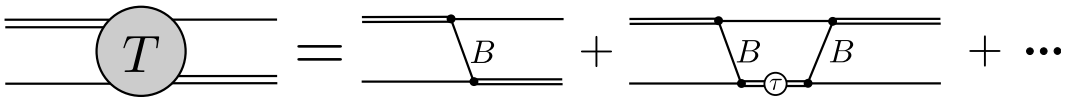


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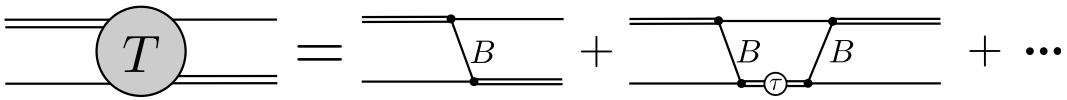


- But, we need the amplitudes for $q \in \mathbb{R} \rightarrow$ can be done in different ways!

- The solution of the Bethe-Salpeter equation can be given in form of the Born series.



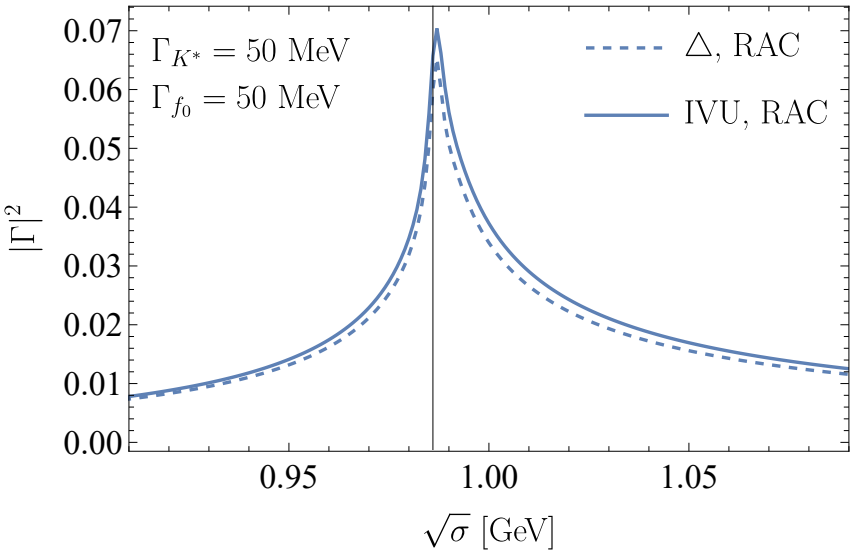
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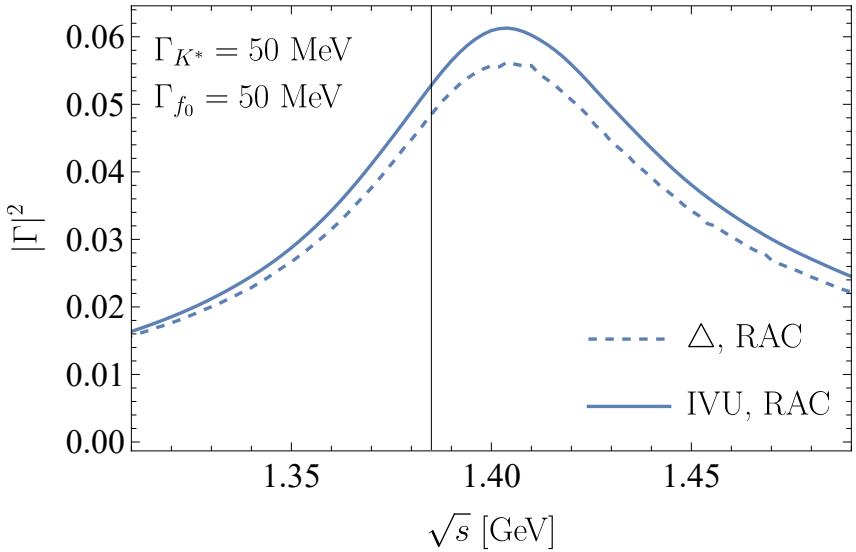
- This defines the loop expansion for the invariant mass distributions.

- **Rational Analytic Continuation**: We use a continued fraction to obtain the isobar-spectator amplitude for $q \in \mathbb{R}$.

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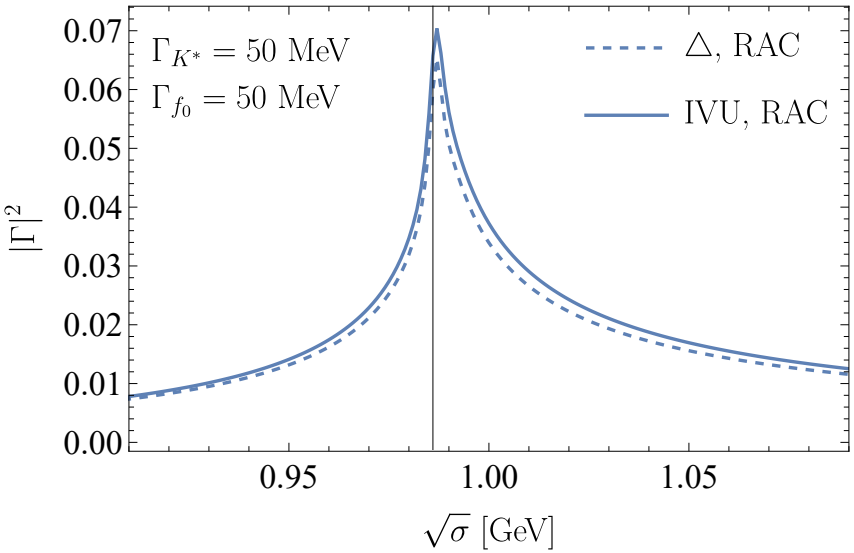


Amplitude squared vs. $\sqrt{\sigma}$, for $\sqrt{s} = 1.42$ GeV.

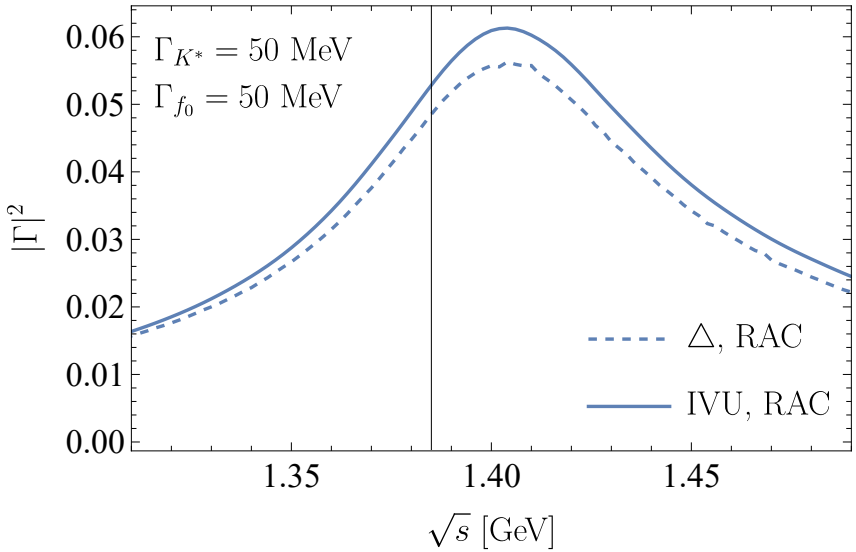


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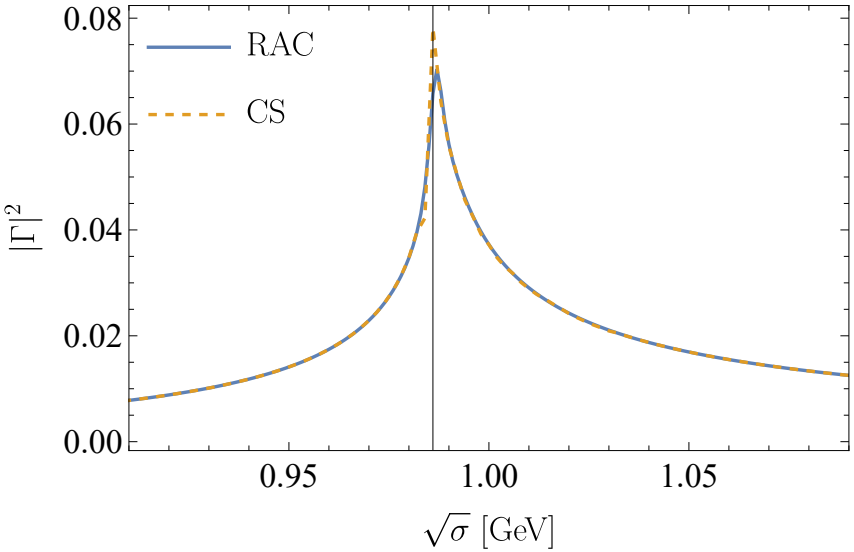


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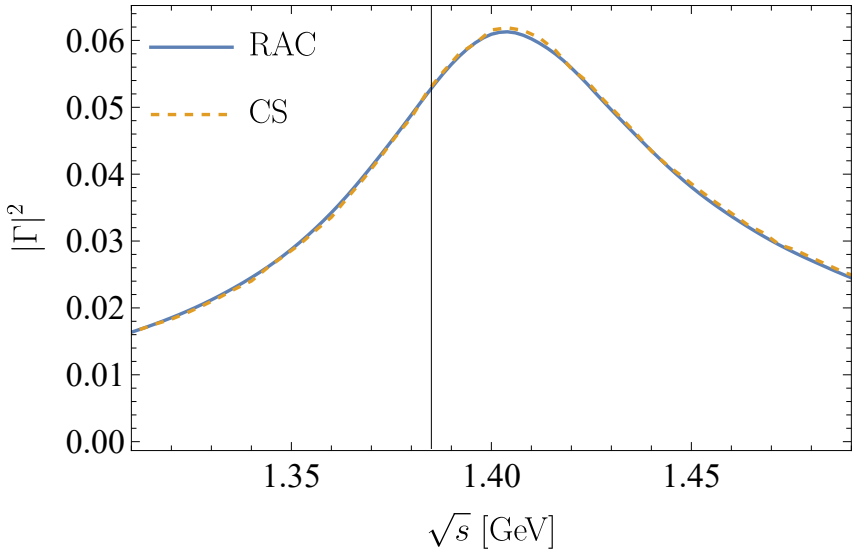
- We see the Landau singularity in both the plots.

- **Cahill & Sloan Method**: We make use Cauchy's integral theorem instead.

- **Cahill & Sloan Method**: We make use Cauchy's integral theorem instead.
- Matches with the other method.



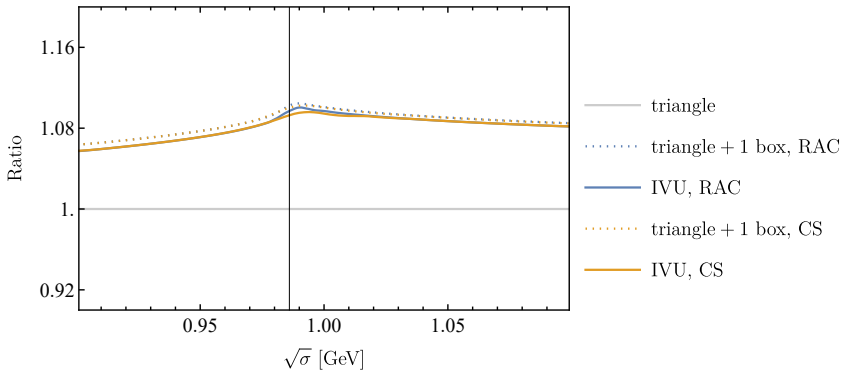
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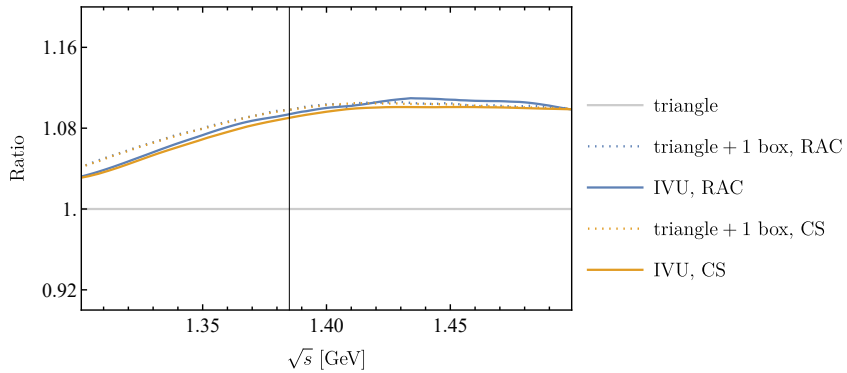
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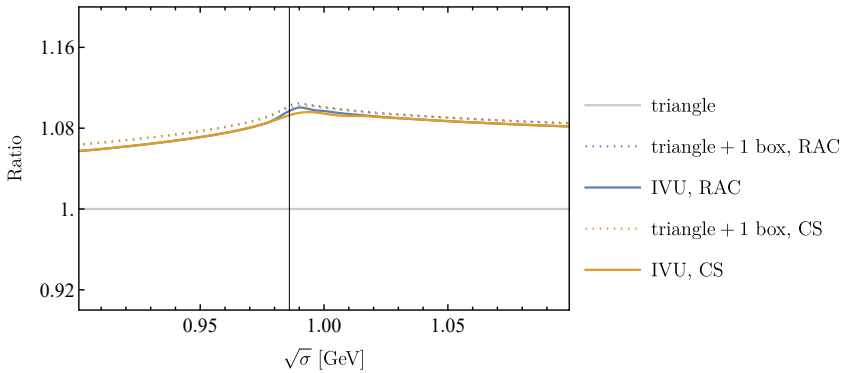


Ratios of amplitudes to triangle amplitudes vs. $\sqrt{\sigma}$, for $\sqrt{s} = 1.42$ GeV.

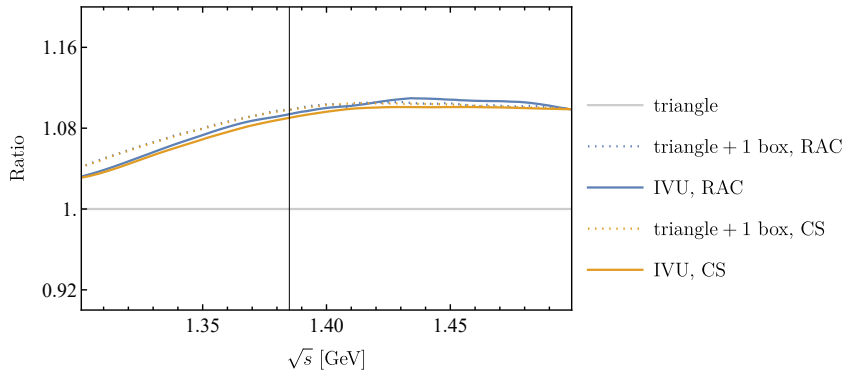


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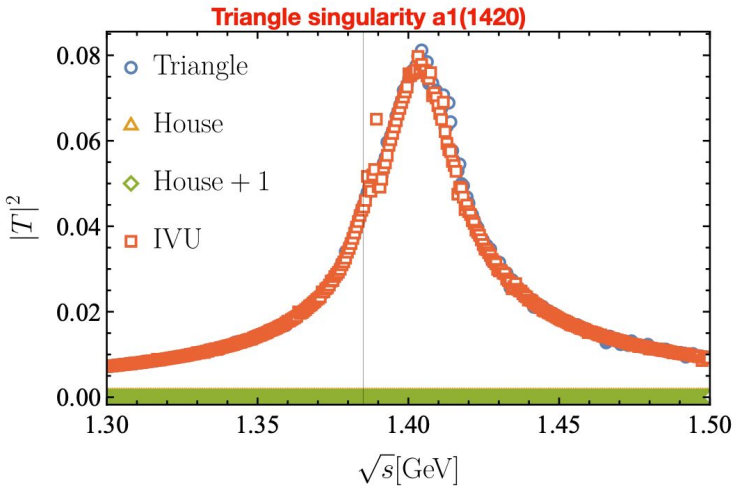
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Ratios of amplitudes to triangle amplitudes vs. $\sqrt{s}$, for $\sqrt{\sigma} = 0.99$ GeV.

- The amplitudes at triangle +1 box are about 1.1x the triangle diagram amplitudes.

- The amplitudes are narrower and pronounced for unphysical widths.



Sakthivasan/MM/... JHEP 10 (2024) 246
see also Mikhashenko,Ketzer, ...

Extensions

- This implementation can be extended to include spin and fit to experimental results (Feng et al., Phys. Rev. D. 110(2024)) — in progress.

Unitary coupled-channel three-body amplitude with pions and kaons

Yuchuan Feng,[1,*] Fernando Gil,[2,†] Michael Döring,[1,3,‡] Raquel Molina,[2,§] Maxim Mai,[1,4,5] Vanamali Shastry,[6,7] and Adam Szczepaniak[6,7,3]

[1]*The George Washington University, Washington, DC 20052, USA*
[2]*IFIC (CSIC-UV, Catedrático José Beltrán Martinez 2, Valencia, Spain)*
[3]*Theory Center, Thomas Jefferson National Accelerator Facility, Newport News, Virginia 23606, USA*
[4]*Albert Einstein Center for Fundamental Physics, Institute for Theoretical Physics, University of Bern, Sidlerstrasse 5, 3012 Bern, Switzerland*
[5]*Helmholtz-Institut für Strahlen- und Kernphysik (Theorie) and Bethe Center for Theoretical Physics, Universität Bonn, 53115 Bonn, Germany*
[6]*Department of Physics, Indiana University, Bloomington, Indiana 47405, USA*
[7]*Center for Exploration of Energy and Matter, Indiana University, Bloomington, Indiana 47403, USA*

(Received 23 July 2024; accepted 6 September 2024; published 4 November 2024)

Three-body dynamics above threshold is required for the reliable extraction of many amplitudes and resonances from experiment and lattice QCD. The S-matrix principle of unitarity can be used to construct dynamical coupled-channel approaches in which three particles scatter off each other, rearranging two-body subsystems by particle exchange. This paper reports the development of a three-body coupled-channel, amplitude including pions and kaons. The unequal-mass amplitude contains two-body S- and P-wave subsystems ("isobars") of all isospins, $I = 0, 1/2, 1, 3/2, 2$, and it also allows for transitions within a given isobar. The $f_0(500)(\sigma), f_0(980), \rho(700), K_0^*(700)(\kappa)$, and $K^*(892)$ resonances are included, apart from repulsive isobars. Different methods to evaluate the amplitude for physical momenta are discussed. Production amplitudes for a_1 quantum numbers are shown as a proof of principle for the numerical implementation.

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- Preliminary results show that the triangle singularity is still present at $1 + \infty$-loop level.

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- Triangle singularity at 1-loop level and at higher loops as subleading singularity.

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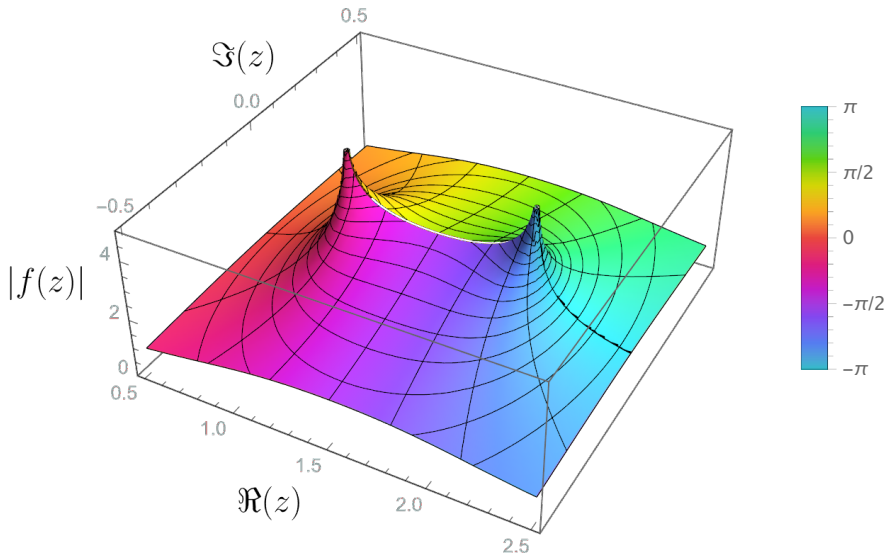
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- Outlook:
  - $a_1(1420)$ from experimental data? Id est, can we resolve the question of a genuine resonance from COMPASS data? — in progress.

Thanks for listening!

Landau singularities

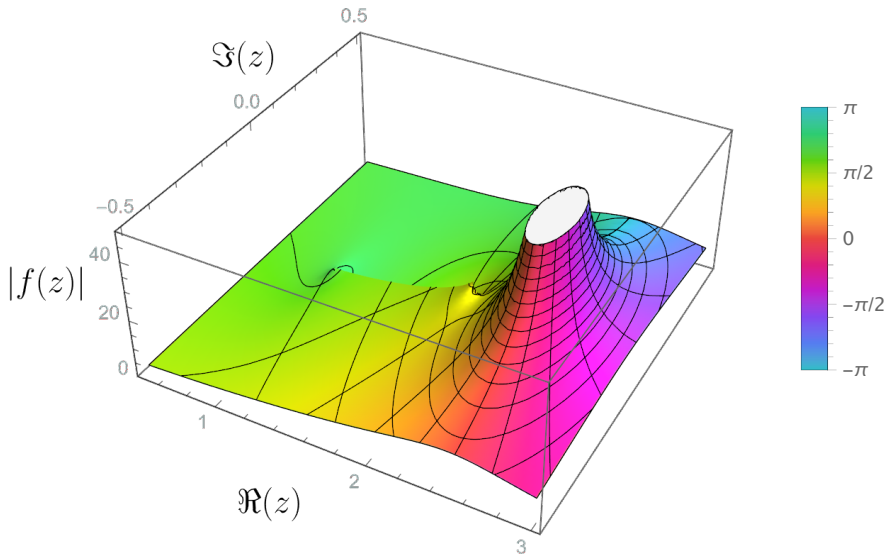
Endpoint singularities: singularities in g, hitting the contour C. E.g.,

$$f(z) = \int_1^2 dw \, \frac{1}{w - z}. \qquad (1)$$

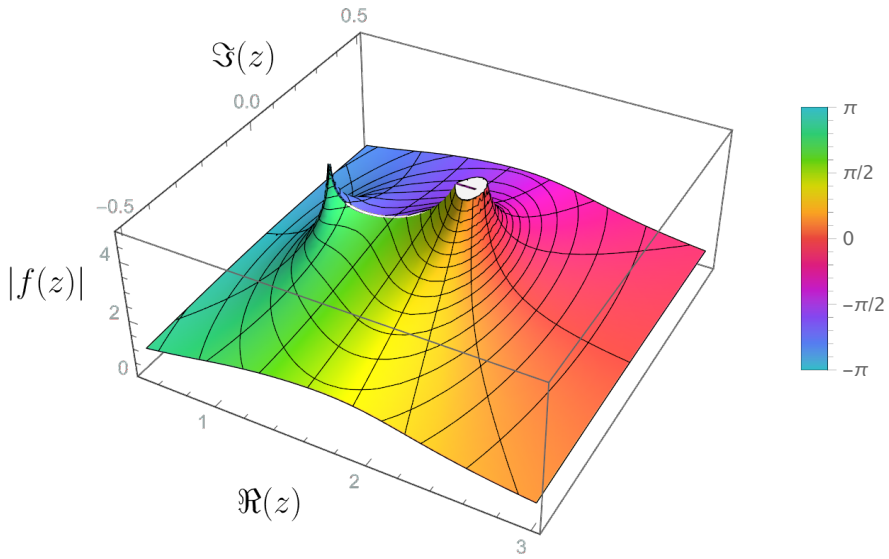


Pinch singularities: Contour C, gets trapped between two singularities in g. E.g.,

$$f(z) = \int_1^2 dw \; \frac{1}{(w-z)(w-5/2)}. \tag{2}$$



Pinch singularity avoided when the singularities approach from the same side of the contour.



The Other Method

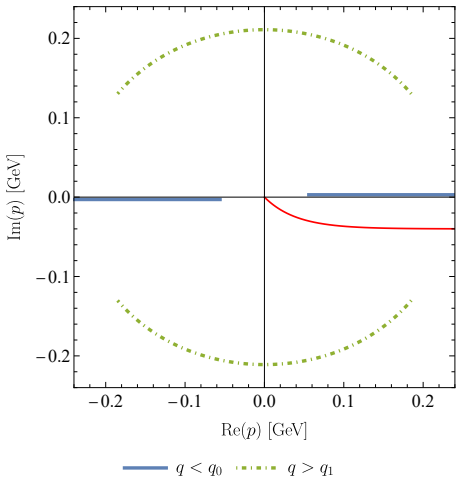
- **Cahill & Sloan Method**[5]: Instead of a continued fraction, use the knowledge of the location of the branch cuts to analytically continue the amplitude for $q \in \mathbb{R}$.[6]

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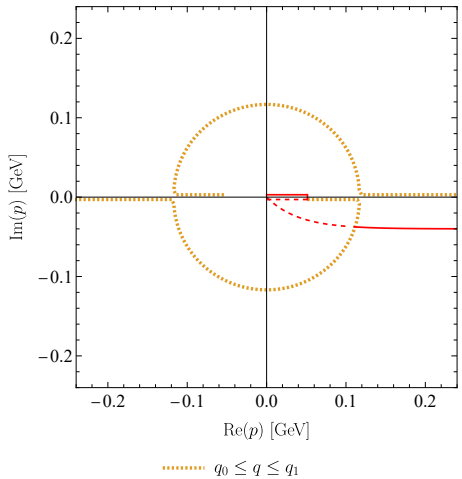
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Branch cuts of B for q outside the triangle singularity region.



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